

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-3

Presentation Time: 1:00 PM-5:00 PM

ENHANCED FORECASTING OF FLOOD EVENTS BY INTEGRATING ARIMA IN TIME SERIES DECOMPOSITION: AN APPLICATION AT THE DELAWARE RIVER, TRENTON, NJ

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Flood forecasting is crucial to saving billions of dollars by fostering the resilience of a society and saving countless lives. Heavy rainfall and warmer temperatures escalate the amount of water discharge and the probability of ice jams in the Delaware River, this intensifies Trenton, New Jersey's risk of flooding. Although it is difficult to predict short-term outcomes, this research utilized time series analysis and multiple autoregressive integrated moving average (ARIMA) for flood predictions in Trenton, New Jersey. Time series decomposition has been applied to separate the noise from the signal of available hydrogeological and climatic data from the time span of April of 2005 to August of 2018. Data includes precipitation, wind speed, wind direction, air temperature, water temperature, tides, depth to groundwater, and water discharge. Time series were separated into different time scale components such as the long-term, seasonal-term, and short-term utilizing the Kolmogorov-Zurbenko filter. Lag operators were applied to the time series decomposition using a data mining software, KNIME, to maximize the performance model of both hydrogeological and climatic fluctuations. ARIMA was implemented to both the raw data and the difficult to predict short-term component to enhance the overall model and strengthen the association between water discharge and hydrogeological data as well as climatic data. The raw data modeling with a multiple linear regression model (MLR) provides a total explanation of 54%, while the application of the Time Series Regression (TSR) model and the ARIMA model in the decomposed data provides a total explanation up to 89%. This methodology can be applied for the prediction of the water discharge at multiple sites.

Session No. 16--Booth# 3

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

[Carolina Ballroom \(Francis Marion Hotel\)](#)

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[Back to: T12. Natural Hazards Prediction \(Posters\)](#)

[<< Previous Abstract](#) | [Next Abstract >>](#)